

jakobian determinant

Jakobian (Jacobian) i jakobian determinant su osnovni pojmovi iz višedimenzionalnog računa i ML-a.

1) Jakobian (Jacobian)

Za vektorsku funkciju

$$\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \mathbf{f}(x_1, \dots, x_n) = \begin{bmatrix} f_1(x_1, \dots, x_n) \\ \vdots \\ f_m(x_1, \dots, x_n) \end{bmatrix},$$

jakobian je matrica parcijalnih izvoda:

$$J_{\mathbf{f}}(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

- Dimenzije: $m \times n$
- Intuicija: lokalna linearna aproksimacija transformacije oko tačke x

2) Jakobian determinant

Ako je $m = n$ (ista dimenzija ulaza i izlaza), možemo uzeti **determinantu jakobiana**:

$$\det(J_{\mathbf{f}}(x)).$$

Značenje:

- $|\det J|$ = faktor lokalne promjene zapremine (ili površine)
- Znak:
 - > 0 : orijentacija očuvana
 - < 0 : orijentacija "okrenuta"

- $= 0$: transformacija "sabija" dimenziju (nije lokalno invertibilna)

3) Kratki primjeri

(a) 2D $\rightarrow$ 2D

$$f(x, y) = (x^2 + y, x - y)$$

$$J = \begin{bmatrix} 2x & 1 \\ 1 & -1 \end{bmatrix}, \quad \det J = -2x - 1.$$

(b) Promjena koordinata (polarno)

$$(x, y) = (r \cos \theta, r \sin \theta)$$

$$\det J = r$$

$\rightarrow$ zato se u integralu pojavljuje faktor $r \, dr \, d\theta$.

4) Zašto je bitno (posebno u ML-u)?

- **Backpropagation**: gradijenti su jakobiani složenih funkcija
- **Normalizing flows**: log-determinanta jakobiana u gustoćama
- **Invertibilnost**: $\det J \neq 0 \Rightarrow$ lokalno invertibilna mapa
- **Geometrija**: razumijevanje deformacije prostora pod mapom

Ako želiš, mogu:

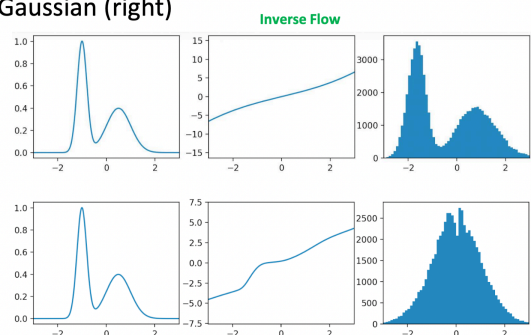
- riješiti **konkretni zadatak** korak-po-korak
- povezati sa **gradijentom vs. jakobianom** (česta konfuzija)
- objasniti **lančano pravilo za jakobiane** (kompozicija funkcija)



Flows: Example

- Inverse flow to a Gaussian (right)

- Before training:



- After training:

explain each graph:

Let's go **panel by panel**, left $\rightarrow$ right, and **top row (before training)** vs **bottom row (after training)**.

This is a **1D normalizing flow** example.

Big picture (what this slide shows)

- We have **data distribution** (left plots): non-Gaussian, bimodal.
- We learn an **inverse flow** f^{-1} that maps data $x \rightarrow$ latent z .
- Goal: after training, the mapped latent variables should look like a **standard Gaussian**.
- The **Jacobian determinant** of the inverse flow is what makes the density change correctly.

BEFORE TRAINING (top row)

Top-left: Data distribution $p_X(x)$

- This is the **original data density**
- Clearly **non-Gaussian** (two bumps / modes)
- This is what we want the model to learn

➡ This distribution is *fixed* before and after training.

Top-middle: Inverse flow function $z = f^{-1}(x)$

- This is the **mapping function**
- Before training:
 - Almost linear
 - No strong curvature
 - Not expressive enough to “unfold” the bimodal structure

➡ Poor mapping $\Rightarrow$ density won't become Gaussian.

Top-right: Latent distribution $p_Z(z)$

- Take data samples x
- Push them through inverse flow: $z = f^{-1}(x)$
- Histogram of resulting z

Before training:

- Still **bimodal**
- Definitely **not Gaussian**

➡ Flow has learned nothing yet.

AFTER TRAINING (bottom row)

Bottom-left: Data distribution $p_X(x)$

- Same as before
 - The data **never changes**
 - Only the transformation changes
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Bottom-middle: Learned inverse flow $z = f^{-1}(x)$

- Now the function is **non-linear**
- Notice:
 - Steeper slopes where density is high
 - Flatter where density is low

- This reshapes space so mass spreads evenly

➡ This nonlinearity is what the **Jacobian determinant** captures.

Bottom-right: Latent distribution $p_Z(z)$

- After applying the trained inverse flow:
 - Distribution is now **approximately Gaussian**
 - Symmetric
 - Single mode

➡ **Success:**

$$p_Z(z) \approx \mathcal{N}(0, 1)$$

Where the Jacobian determinant fits in

For inverse flow $z = f^{-1}(x)$:

$$\log p_X(x) = \log p_Z(z) + \log \left| \det \frac{dz}{dx} \right|$$

- The **middle plot** controls $\frac{dz}{dx}$
- Its slope = Jacobian
- Steep slope → stretches space → lowers density
- Flat slope → compresses space → increases density

One-sentence summary per column

Column	Meaning
Left	Original data density (unchanged)
Middle	Learned inverse mapping (geometry of space)
Right	Resulting latent density

If you want, I can:

- derive the **change-of-variables formula** step-by-step
- explain why **inverse flow** (not forward) is trained
- connect this directly to **log-det-Jacobian in code** (RealNVP / Glow)



$$\max_{\theta} \underbrace{\sum_i \log(p_x(x^{(i)}; \theta))}_{\text{Maximum Likelihood}} = \max_{\theta} \left(\underbrace{\sum_i \log(p_z(f_{\theta}^{-1}(x^{(i)}))}_{\text{Latent variable version}} + \log \underbrace{\left| \frac{\partial f_{\theta}^{-1}(x^{(i)})}{\partial x} \right|}_{\text{Determinant of Jacobian matrix}} \right)$$

can you give me a sample exercise i might get for this formula or overall for Flow-Based Models

Great question — this is **exactly** the kind of thing instructors like to test for **flow-based models / change-of-variables**. Below are **realistic exam-style exercises**, ordered from **easy → harder**, all directly using the formula you showed.

◆ Exercise 1 (Warm-up: change of variables, 1D)

Let

$$z = f^{-1}(x) = 2x + 1$$

Assume the latent variable

$$z \sim \mathcal{N}(0, 1)$$

(a)

Compute $p_X(x)$.

(b)

Write $\log p_X(x)$ explicitly.

(c)

Is this a valid normalizing flow? Why?

✓ Expected reasoning

- Compute Jacobian:

$$\left| \frac{dz}{dx} \right| = 2$$

- Apply change of variables:

$$p_X(x) = p_Z(2x + 1) \cdot 2$$

◆ Exercise 2 (Core exam question – log-likelihood)

You are given a **1D normalizing flow**:

$$z = f_{\theta}^{-1}(x) = x + \theta \sin x$$

with base distribution $z \sim \mathcal{N}(0, 1)$.

(a)

Write the expression for the log-likelihood

$$\log p_X(x; \theta)$$

using the change-of-variables formula.

(b)

Compute the Jacobian term

$$\log \left| \frac{dz}{dx} \right|$$

(c)

State **one condition on θ** that ensures the flow is invertible.

✓ Expected steps

$$\frac{dz}{dx} = 1 + \theta \cos x$$

Invertibility condition:

$$1 + \theta \cos x \neq 0 \quad \forall x \Rightarrow |\theta| < 1$$

◆ Exercise 3 (Maximum Likelihood Training – very common)

You are given i.i.d. data points

$$\{x^{(i)}\}_{i=1}^N$$

The flow is:

$$z^{(i)} = f_{\theta}^{-1}(x^{(i)}) = \theta x^{(i)}$$

and

$$z \sim \mathcal{N}(0, 1)$$

(a)

Write the full **log-likelihood objective**

$$\sum_i \log p_X(x^{(i)}; \theta)$$

(b)

Explain qualitatively what happens to θ during training.

✓ Key insight

$$\log p_X(x) = -\frac{1}{2}(\theta x)^2 + \log |\theta| + \text{const}$$

Training balances:

- data fit term $(\theta x)^2$
- volume correction $\log |\theta|$

◆ Exercise 4 (2D Jacobian – very exam-relevant)

Consider a 2D flow:

$$\begin{cases} z_1 = x_1 \\ z_2 = x_2 e^{s(x_1)} \end{cases}$$

(a)

Write the Jacobian matrix $J = \frac{\partial z}{\partial x}$

(b)

Compute $\det J$

(c)

Explain why this form is used in RealNVP / Glow.

✓ Solution sketch

$$J = \begin{bmatrix} 1 & 0 \\ x_2 e^{s(x_1)} s'(x_1) & e^{s(x_1)} \end{bmatrix}$$

$$\det J = e^{s(x_1)}$$

➡ Easy log-determinant, guaranteed invertibility.

◆ Exercise 5 (Conceptual – guaranteed to appear)

Answer briefly:

1. Why do flow models maximize **exact likelihood**, unlike VAEs?
2. Why is the Jacobian determinant needed?

3. What happens if the Jacobian determinant is zero?
 4. Why do flows often train the **inverse** mapping?
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✓ Expected answers (short)

1. Exact density via change-of-variables
 2. Corrects for volume distortion
 3. Mapping collapses space → invalid density
 4. Inverse is cheap to evaluate likelihood
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◆ Exercise 6 (Hard / oral-exam style)

You train a flow and observe:

- The latent distribution is Gaussian
- The Jacobian determinant is constant

Question:

What class of transformations has the model learned?

✓ **Answer:** affine / linear transformations only.

🔑 If you want

I can:

- give you **fully worked solutions** to any exercise
- generate a **mock midterm problem** (CS760 / ML style)
- connect this formula directly to **RealNVP / Glow code**

Just tell me which one 👍



↳ ****fully worked solutions**** to any exercise

ok

Perfect 👍

I'll give you **fully worked, step-by-step solutions** to a **representative exam set** for **Flow-Based Models**, explicitly using the formula you showed.

I'll do **Exercises 1–4**, since those cover **mechanics + intuition + Jacobians**.

✅ Exercise 1 — 1D Change of Variables (fully solved)

Given

$$z = f^{-1}(x) = 2x + 1, \quad z \sim \mathcal{N}(0, 1)$$

(a) Compute $p_X(x)$

Change-of-variables formula (1D):

$$p_X(x) = p_Z(z) \left| \frac{dz}{dx} \right|$$

Compute Jacobian:

$$\frac{dz}{dx} = 2$$

So:

$$p_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(2x+1)^2}{2}} \cdot 2$$

(b) Log-likelihood

$$\log p_X(x) = -\frac{(2x+1)^2}{2} + \log 2 - \frac{1}{2} \log(2\pi)$$

(c) Is this a valid flow?

✓ Yes, because:

- mapping is invertible
- Jacobian determinant $\neq 0$ everywhere
- density integrates to 1

✓ Exercise 2 — Nonlinear Flow + Jacobian

Given

$$z = f_{\theta}^{-1}(x) = x + \theta \sin x, \quad z \sim \mathcal{N}(0, 1)$$

(a) Log-likelihood expression

$$\log p_X(x; \theta) = \log p_Z(z) + \log \left| \frac{dz}{dx} \right|$$

(b) Jacobian term

$$\frac{dz}{dx} = 1 + \theta \cos x$$

So:

$$\log \left| \frac{dz}{dx} \right| = \log |1 + \theta \cos x|$$

(c) Invertibility condition

We must avoid zero Jacobian:

$$1 + \theta \cos x \neq 0 \quad \forall x$$

Since $\cos x \in [-1, 1]$:

$$|\theta| < 1$$

✓ **Key exam insight:** Jacobian $\neq 0 \Rightarrow$ valid flow.

✓ Exercise 3 — Maximum Likelihood Training (classic)

Given

$$z^{(i)} = f_{\theta}^{-1}(x^{(i)}) = \theta x^{(i)} \quad , \quad z \sim \mathcal{N}(0, 1)$$

(a) Full log-likelihood objective

Jacobian:

$$\frac{dz}{dx} = \theta$$

Log-density:

$$\log p_X(x^{(i)}) = \log p_Z(\theta x^{(i)}) + \log |\theta|$$

Expand:

$$= -\frac{1}{2}(\theta x^{(i)})^2 + \log |\theta| - \frac{1}{2} \log(2\pi)$$

Summing over data:

$$\sum_i \log p_X(x^{(i)}) = -\frac{\theta^2}{2} \sum_i (x^{(i)})^2 + N \log |\theta| + \text{const}$$

(b) What does training do?

- First term penalizes large θ
- Second term encourages large θ
- ML finds a **balance** matching data variance

✓ This is why flows **learn scale automatically**

✓ Exercise 4 — 2D Flow (RealNVP-style)

Given

$$\begin{cases} z_1 = x_1 \\ z_2 = x_2 e^{s(x_1)} \end{cases}$$

(a) Jacobian matrix

$$J = \frac{\partial z}{\partial x} = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ x_2 e^{s(x_1)} s'(x_1) & e^{s(x_1)} \end{bmatrix}$$

(b) Determinant

$$\det J = 1 \cdot e^{s(x_1)} = e^{s(x_1)}$$

So:

$$\log |\det J| = s(x_1)$$

(c) Why this is used in RealNVP / Glow

✓ Because:

- determinant is **cheap**
 - invertibility guaranteed
 - expressive nonlinear mapping
 - efficient likelihood computation
-

Master takeaway (what profs want)

If you can:

- write **log-likelihood**
- compute **Jacobian determinant**
- state **invertibility conditions**
- explain **why flows work**

👉 you will ace any flow-based models question.

If you want next:

- 🔥 **full mock midterm problem** (with grading rubric)
- 🗃️ **multi-layer flow (composition of Jacobians)**
- 💻 **map each term to RealNVP / Glow code**

Just tell me.

